

Task-1

To set up a basic wireless network using a wireless router and connect a laptop to the network.
The SSID and default wireless channel are recorded.



Parameter	Configuration
Device	Wireless Router
SSID	MyHomeWiFi
Wireless Channel	6
Connected Device	Laptop
Connection Type	Wireless
IP Address	192.168.0.100
Default Gateway	192.168.0.1
Status	Connected

- Open **Cisco Packet Tracer**.
- Add a **Wireless Router** from **Network Devices** → **Wireless Devices**.
- Add a **Laptop** from **End Devices**.
- Open the Wireless Router and go to **GUI** → **Wireless** → **Basic Wireless Settings**.
- Set the **SSID** to MyHomeWiFi.
- Set the **Channel** to 6.
- Save the settings.
- Open the Laptop → **Desktop** → **PC Wireless**.
- Select MyHomeWiFi and connect to it.
- Open **Desktop** → **Command Prompt** and use ipconfig to verify the IP address. Use ping 192.168.0.1 to verify connectivity.

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Cisco Packet Tracer PC Command Line 1.0
C:\>
ipconfig

Bluetooth Connection:(default port)

Connection-specific DNS Suffix.:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
                                0.0.0.0

Wireless0 Connection:

Connection-specific DNS Suffix.:
Link-local IPv6 Address.....: FE80::207:ECFF:FE88:C1EE
IPv6 Address.....: ::
IPv4 Address.....: 192.168.0.100
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
                                192.168.0.1

C:\>

```

Task -2

Wi-Fi Channel Interference Test

Steps

1. **Open Router Settings**
 - Connect your phone/laptop to your Wi-Fi.
 - Open a browser and enter the router's gateway address, such as 192.168.1.1 or 192.168.0.1.
 - Log in with the router administrator credentials.
2. **Change the Wi-Fi Channel**
 - Go to **Wireless/Wi-Fi Settings**.
 - Find **Channel** or **Wi-Fi Channel**.
 - Change it from **Auto** to **1, 6, or 11**.
 - Save the settings and reconnect to Wi-Fi.
3. **Check Nearby Networks**
 - Open a Wi-Fi analyzer app on your phone.
 - Scan the nearby Wi-Fi networks.
 - Check which channels are being used and look for overlapping networks.
4. **Select the Better Channel**
 - Compare channels **1, 6, and 11**.
 - Choose the channel with fewer nearby networks/less overlap.
 - If necessary, change the router channel again and rescan.

Observation Table

Channel	Nearby Networks	Interference
1	4	High
6	1	Low
11	3	Medium

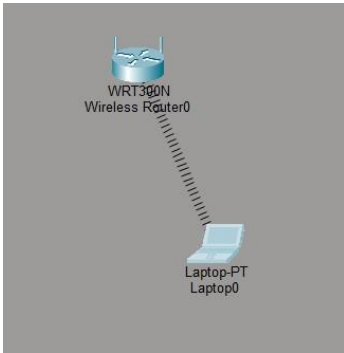
Example Result: Channel 6 had the least interference, so it was selected manually instead of Auto.

Conclusion: Manually selecting a less crowded Wi-Fi channel can reduce interference and may improve Wi-Fi stability and performance.

Task-3

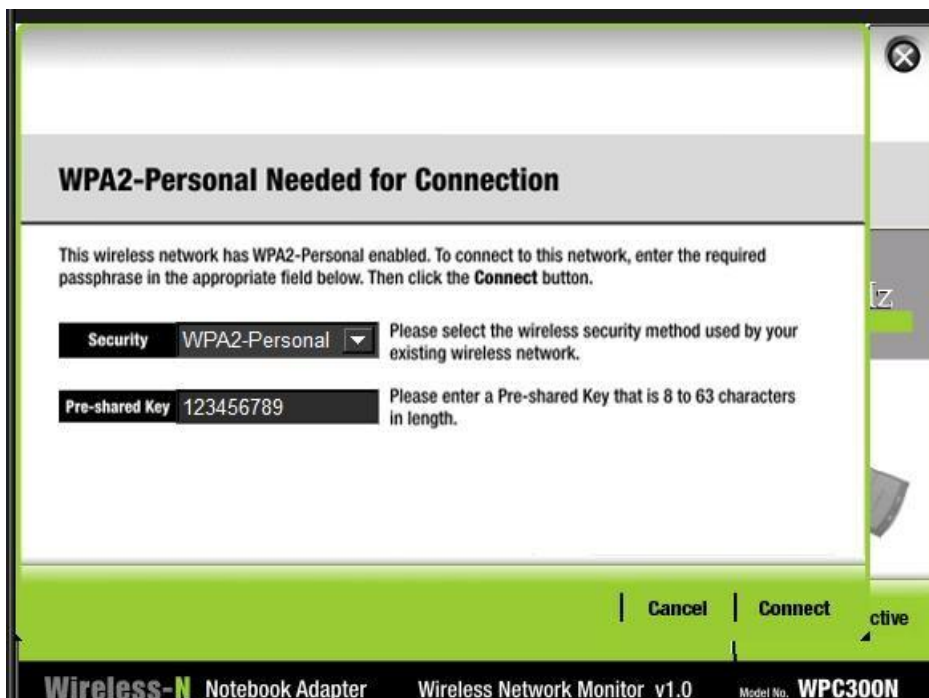
Q- Enable WPA2 or WPA3 security on your wireless router, set a strong password, and attempt to connect a device using the new credentials to verify the security setting is active.

To enable **WPA2 security** on a wireless router, configure a strong password, and connect a device using the new credentials to verify that wireless security is active.



Parameter	Configuration
Device	Wireless Router
SSID	MyHomeWiFi
Security Mode	WPA2-Personal
Encryption	AES
Password	123456789
Connected Device	Laptop
Connection Type	Wireless

1. Open **Cisco Packet Tracer**.
2. Add a **Wireless Router** and a **Laptop**.
3. Open the Wireless Router and select **GUI**.
4. Go to **Wireless → Wireless Security**.
5. Select **WPA2-Personal** as the security mode.
6. Select **AES** encryption.
7. Set the password to 123456789
8. Click **Save Settings**.
9. On the Laptop, open **Desktop → PC Wireless**.
10. Select the MyHomeWiFi network.
11. Enter the configured WPA2 password.
12. The laptop successfully connects to the wireless network.
13. Enter an incorrect password to verify that the connection fails.
14. Re-enter the correct password and verify that the laptop connects successfully.
15. Use ping 192.168.0.1 to verify connectivity.



```
Cisco Packet Tracer PC Command Line 1.0
C:\>
ipconfig

Bluetooth Connection:(default port)

Connection-specific DNS Suffix...:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
                                0.0.0.0

Wireless0 Connection:

Connection-specific DNS Suffix...:
Link-local IPv6 Address.....: FE80::207:ECFF:FE88:C1EE
IPv6 Address.....: ::
IPv4 Address.....: 192.168.0.100
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
                                192.168.0.1

C:\>
```

Result

WPA2-Personal security was successfully enabled on the wireless router. A strong password was configured, and the laptop successfully connected using the correct credentials. An incorrect password was rejected, confirming that the wireless security setting is active.

Task-4

MAC Filtering Configuration

Steps

1. Find Your Device's MAC Address

- On Windows, open Command Prompt and type:
ipconfig /all
- Find Physical Address under your Wi-Fi adapter.
- Example: A4-B1-C1-22-33-44
- 2. Open Router Settings**
 - Connect to your Wi-Fi.
 - Open the router's gateway address, such as 192.168.1.1.
 - Log in to the router.
- 3. Enable MAC Filtering**
 - Go to Wireless / Wi-Fi Settings → MAC Filtering / Access Control.
 - Enable MAC filtering.
 - Select Allow/Whitelist mode.
- 4. Add Your Device**
 - Enter your device's MAC address into the allowed list.
 - Save or apply the settings.
- 5. Test Another Device**
 - Try connecting a phone/laptop whose MAC address is not in the allowed list.
 - The device should be blocked from connecting or denied network access.

Observation

Device	MAC Address	Allowed List	Result
My Laptop	A4-B1-C1-22-33-44	Yes	Connected
Other Phone	78-AB-CD-12-34-56	No	Blocked

Conclusion: MAC filtering allows the router to control which devices can access the Wi-Fi network. Only devices whose MAC addresses are in the allowed list can connect.

Task -5

To configure an access control policy on a **WRT300N wireless router** to block access to a specific website, such as www.youtube.com, and verify that the website is blocked from a connected device.

Parameter	Configuration
Router	WRT300N
Access Restriction	Enabled
Policy Name	Block YouTube
Website to Block	youtube.com
Connected Device	Laptop

Connection	Wireless
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Procedure

1. Open **Cisco Packet Tracer**.
2. Add a **WRT300N Wireless Router** and a **Laptop**.
3. Connect the laptop to the WRT300N using its wireless network.
4. Open the **WRT300N → GUI**.
5. Go to **Access Restrictions**.
6. Enable the **Internet Access Policy**.
7. Create a new policy named **Block YouTube**.
8. Select the laptop as the device to which the policy will apply.
9. Under **Website Blocking by URL Address**, enter: youtube.com
10. Save the settings.
11. Open the laptop's **Web Browser**.
12. Try to access:

<http://www.youtube.com>

13. Verify that access to the specified website is blocked.

Result

The **WRT300N wireless router's Access Restriction feature was successfully configured** to block youtube.com. The connected laptop was unable to access the specified website after the restriction was enabled, confirming that the website-blocking policy was active.